UNIVERSITÀ DEGLI STUDI DELLA CAMPANIA "*Luigi Vanvitelli*"
DEPARTMENT OF MATHEMATICS AND PHISICS

Assignment 1

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Course: *Numerical methods for Data Analysis* – Professor: *Prof.essa Rosanna Campagna*
Due date: *April 9th, 2024*

Exercise 1

The file **ModelReductionData.mat** contains a data matrix $X \in \mathbb{R}^{6\times 4000}$. Each column of X represents a data vector in the six dimensional space.

a)*a*)

1. Let's visualize the scatter-plot of one component, that correspond to a row of the X matrix, against the other component. there are $\binom{6}{2}$ different two-dimensional projections since there is no need to plot a component against itself because it will be equal and also, since plotting an *ith* component with a *jth* one is the same as plotting the *jth* component againsth the *ith*, there is no need of plotting the same chart.
   The use of PCA (Principal Component Analysis) as visualisation tool is very interesting and the reason is that PCA allow to visualise centered data in two- or three-dimensional scatter-plot with a the best possible spread. That's because it can be proven that the projection direction that maximizes the spread of the data is given by the first left singular vector, $q = u(1)$:

   $$max\{spread(y)\} = \sigma_1 = \|\mathbf{X^T u^{(1)}}\|$$

   The other projection direction, orthogonal to u_1, that gives the second largest spread is $u^{(2)}$:

   $$max\{spread(\tilde{y})\} = \sigma_2 = \|\tilde{\mathbf{X}}^{\mathbf{T}}\mathbf{u^{(2)}}\|$$

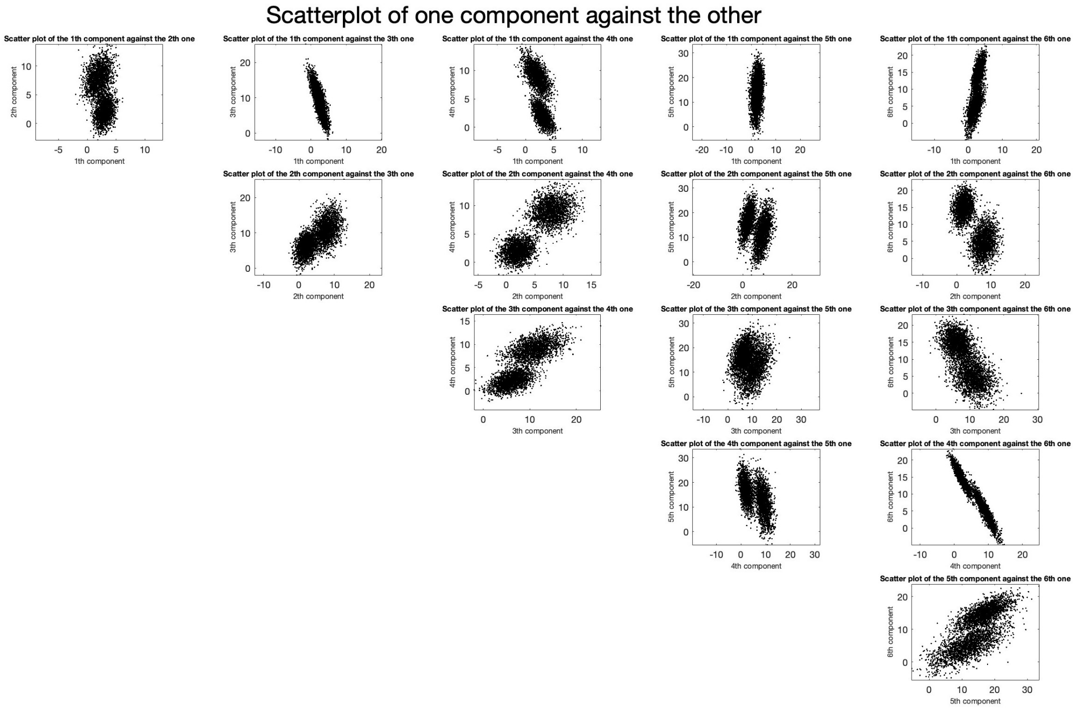


Figure 1: Scatter-plot of the *ith* (with $i = \{1,2,3,4,5,6\}$) against the *jth* one (with $j = \{1,2,3,4,5,6\}$), where i and j represent the rows of the X matrix of the ModelReductionData.mat dataset.

Looking at *Figure 1*, It is easy to notice the presence of cluster in some chart, like in the scatter-plot of the second component against the fourth, and none in some other.

2. After centering the data (firstly, compute the mean value of the data, then, subtract the computed mean from the data vector and store the result in the *centered data matrix* X_c),

$$\bar{x} = \frac{1}{p}\sum_{j=1}^{p} x^{(j)}$$

$$X_c = \begin{bmatrix} x^{(1)} - \bar{x} & x^{(2)} - \bar{x} & ... & x^{(p)} - \bar{x} \end{bmatrix} = \begin{bmatrix} x_c^{(1)} & x_c^{(2)} & ... & x_c^{(p)} \end{bmatrix}$$

I compute the SVD (Singular Value Decomposition):

$$X_c = UDV^T$$

where:
$U \in \mathbb{R}^{6\times6}$ is a matrix with orthonormal columns that contains the left singular vectors
$D \in \mathbb{R}^{6\times4000}$is a diagonal matrix that contains the singular values
$V \in \mathbb{R}^{4000\times4000}$ is a matrix with orthonormal columns that contain the right singular vectors

Plot the singular values (that corresponds to the diagonal values of the D matrix) in a logarithmic scale so that it is possible to visualize them better.

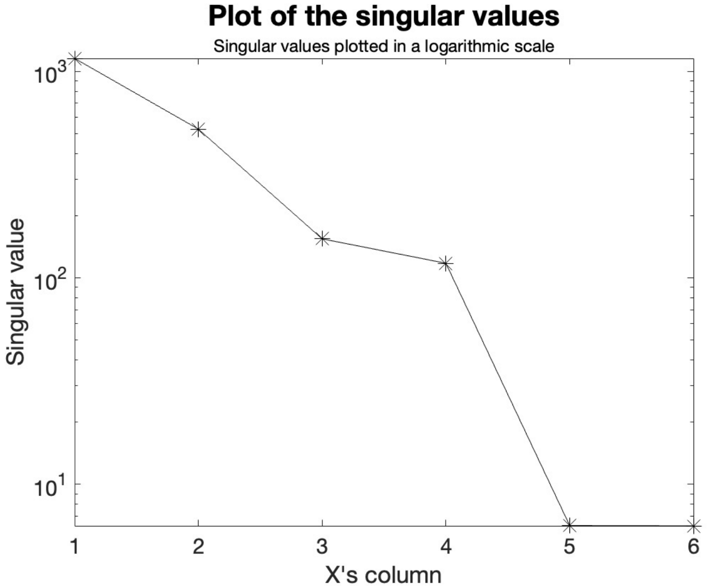


Figure 2: Plot of the singular values σ_j of the diagonal matrix D, result of the SVD (singular value decomposition) of the centered data matrix X_c, in a logarithmic scale.

Since the matrix of *ModelReduction* dataset contains a lot information (4000 columns), the original data can be represented by only a few set of information that allow a big reduction useful both for computational effort and also for eliminate all the noise and capture only the essential information from the column vectors.
From the graph, It's intuitively that we can take only a few singular values in order to approximated the original matrix. In this case, I decide to choose to take only the first 4 feature vectors.

After computing the first k ($k = 4$) principal component of the data,

$$x_c^{(j)} = U_k D_k V_k^T = U_k z^{(j)}$$

$$z^{(j)} = D_k V_k^T e_j$$

$$z^{(j)} = u_k^T x^{(j)}$$

$$Z = U_k^T X$$

we can create the scatter-plots of the first few (4) principal components: there are $\binom{4}{2}$ difference two-dimensional projection.

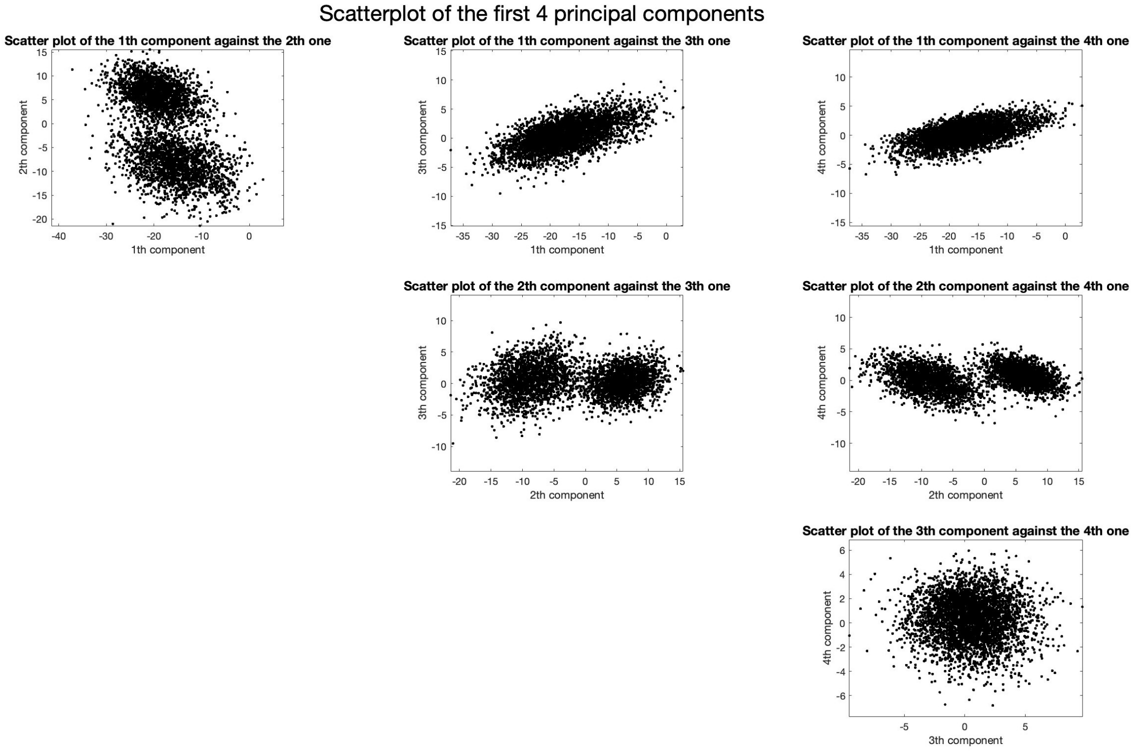


Figure 3: Scatter-plot of the *ith* (with $i = \{1, 2, 3, 4\}$) against the *jth* one (with $j = \{1, 2, 3, 4\}$), where i and j represent the rows of the principal component matrix Z_k derived from applying SVD (Singular Value Decomposition) to the centered data matrix X_c of the ModelReductionData.mat dataset.

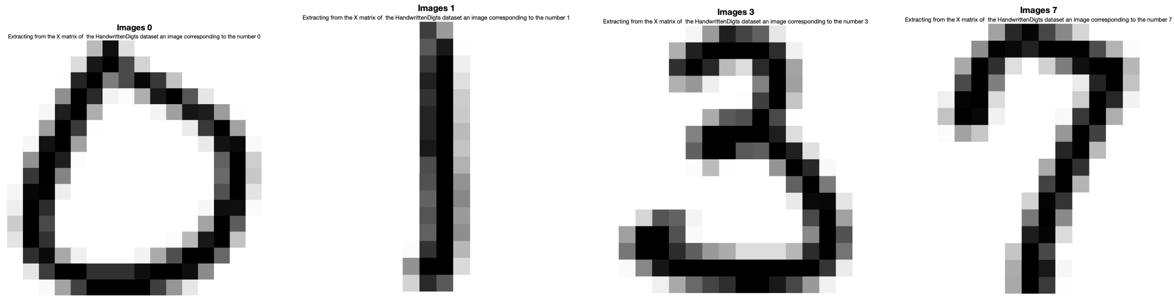
It's interesting to notice from the grid of the *figure 3* the presence of cluster from the scatter-plot of the first component against the second one and the second component against the third and the fourth ones. In the other chart, the points does not suggest any clustering.

Exercise 2

The **HandWrittenDigits.mat** dataset contain the data matrix X and the data matrix I. The data X consist of black and white pixel images of handwritten digits collected from US postal envelopes. An image, strongly down-sampled, consists of 16×16 pixels, with gray scale values that are scaled into the interval $[0, 1]$, the value 1 representing white, and 0 black. The pixel values are listed in lexicographical order and stacked into a vector of length n = $(16)^2 = 256$. Our sample contains $p = 1707$ images, hence the data matrix is $X \in \mathbb{R}^{256 \times 1707}$.
The vector $I \in \mathbb{N}^{707}$ contains a list of indices, or attributes, from 0 to 9, indicating that

$$x^{(j)} \text{ represents the number } I(j) \in \{0, 1, ..., 9\}$$

Extract from X the images that correspond to numbers $0, 1, 3$ and 7, as in *figure 4*.



(a) Number 0. (b) Number 1. (c) Number 3. (d) Number 7.

Figure 4: Extracting from the X matrix of the *HandWrittenDigits* dataset images that correspond to numbers 0, 1, 3 and 7. This was done using the label matrix I.

From each subgroup ($j = \{0, 1, 3, 7\}$), I select 5 samples, as observed in the *figure 5*.

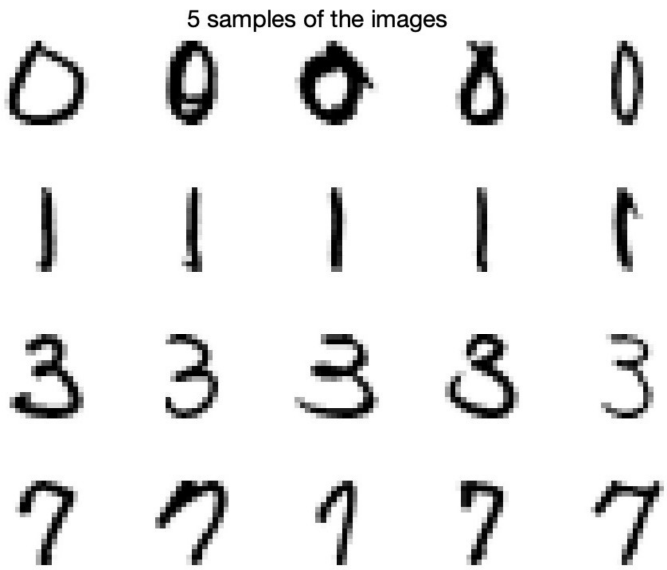


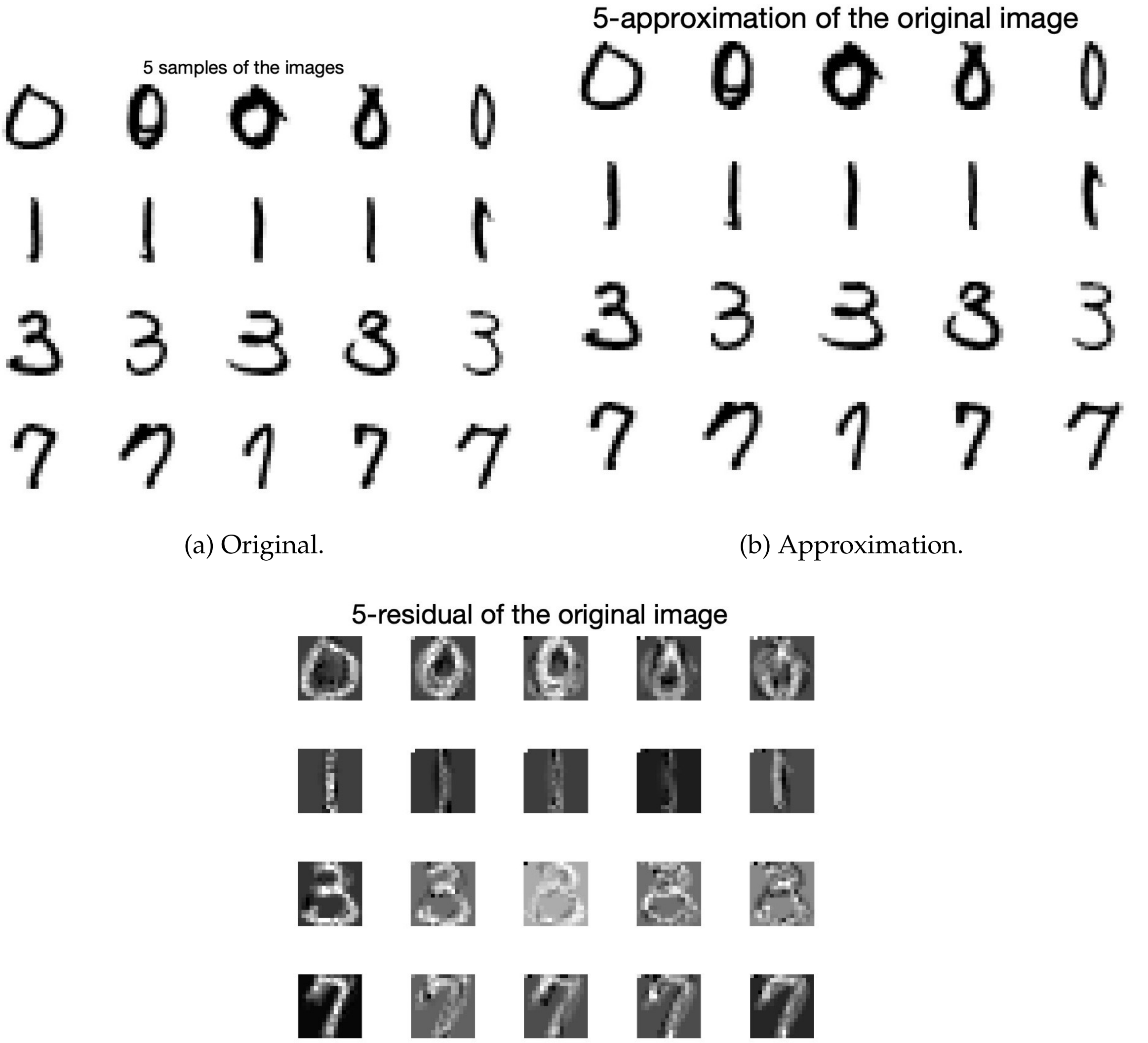
Figure 5: 5 samples of each subgroup of the X matrix of the *HandWrittenDigits* file correspond to numbers 0, 1,3 and 7.

From each 5 samples of each subgroup, I approximate these by the linear combination of the first k feature vectors, $k = 5, 10, 15, 20, 25$, that is,

$$x^{(j)} \approx P_k x^{(j)} = \sum_{l=1}^{k} z_l^{(j)} u^{(l)}$$

where $z^{(j)}$ are the principal components of $x^{(j)}$. I plot the approximation as images, as well as the residual,

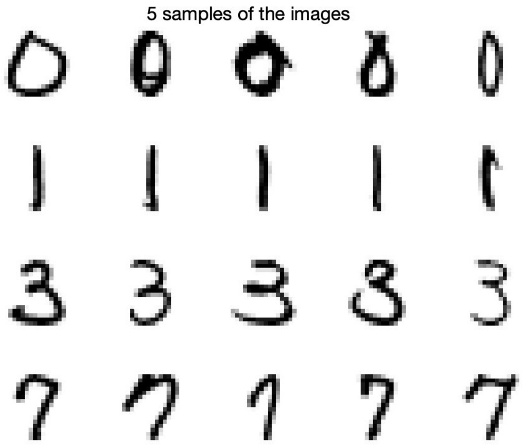
$$x^{(j)} - P_k x^{(j)}$$



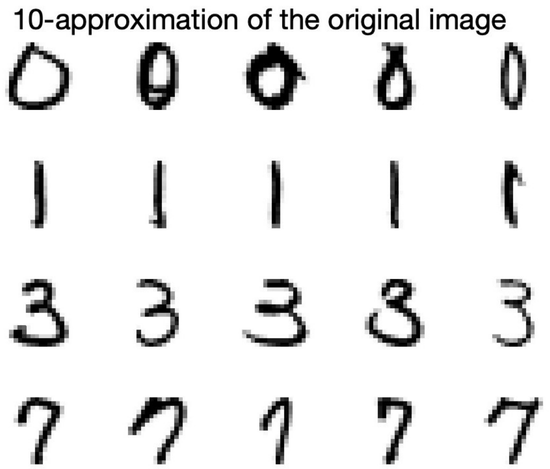
(a) Original. (b) Approximation.

(c) Residual.

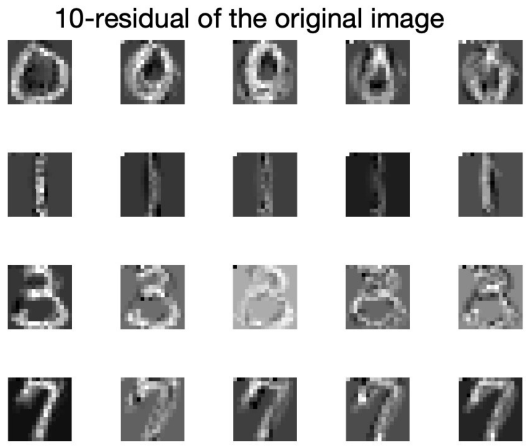
Figure 6: Original data, approximation images and residual between the former and the approximated images by $k = 5$ feature vectors.



(a) Original.

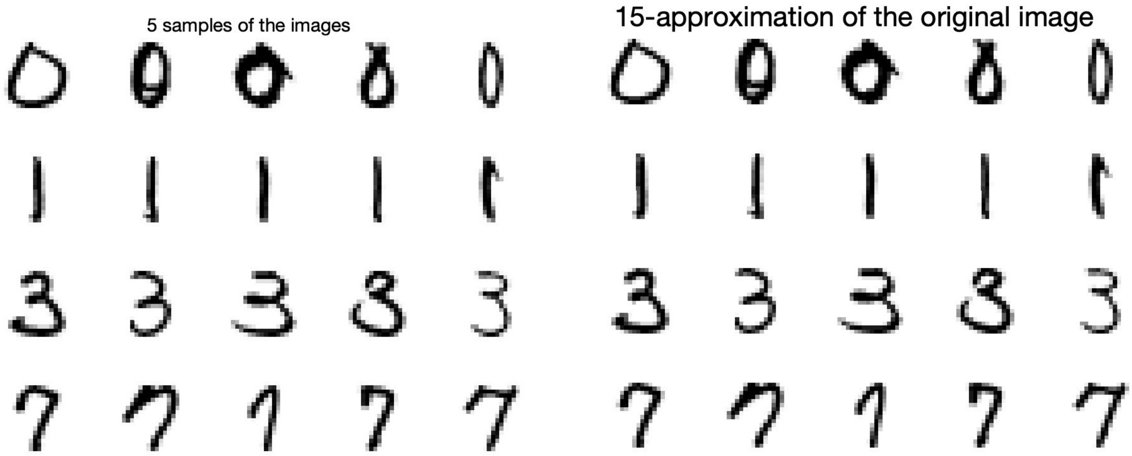


(b) Approximation.

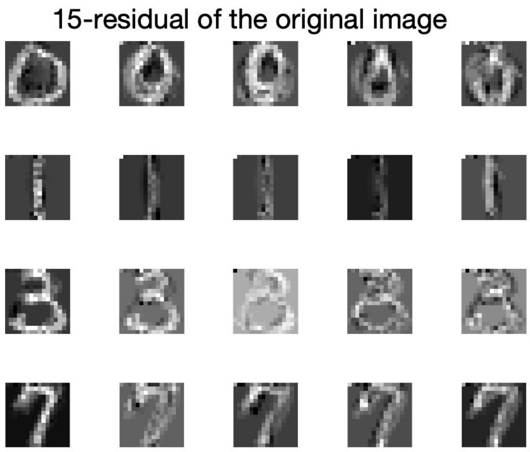


(c) Residual.

Figure 7: Original data, approximation images and residual between the former and the approximated images by $k = 10$ feature vectors.

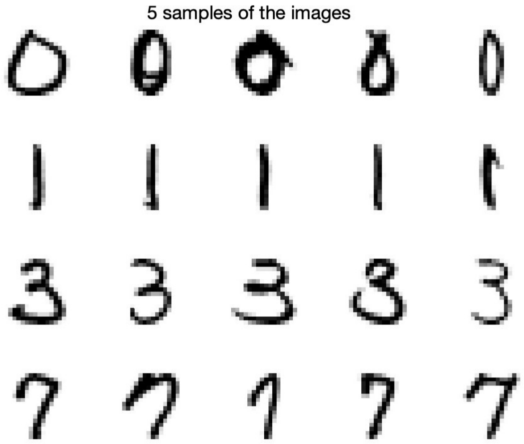


(a) Original. (b) Approximation.

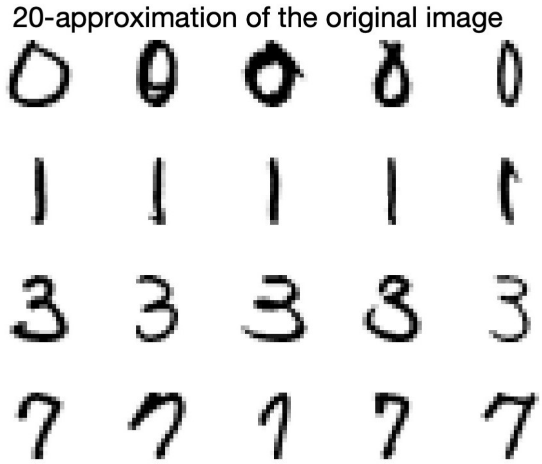


(c) Residual.

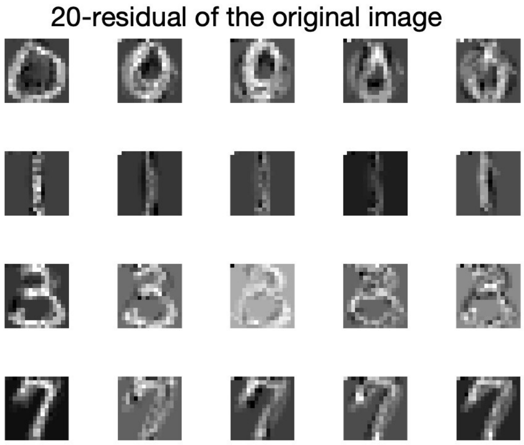
Figure 8: Original data, approximation images and residual between the former and the approximated images by $k = 15$ feature vectors.



(a) Original.

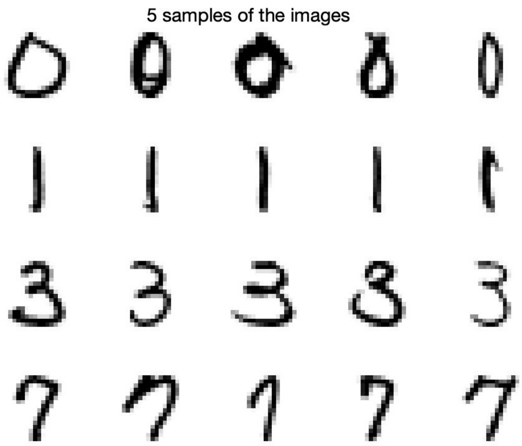


(b) Approximation.

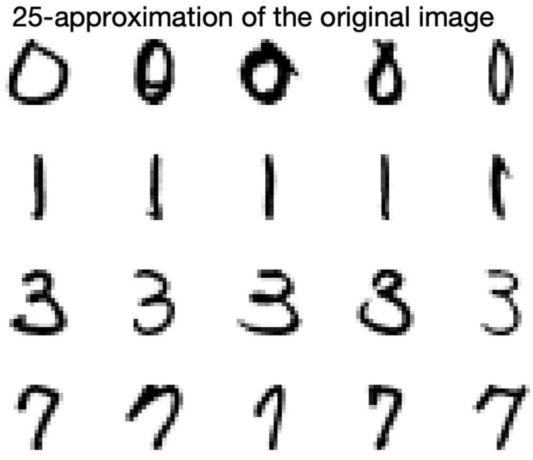


(c) Residual.

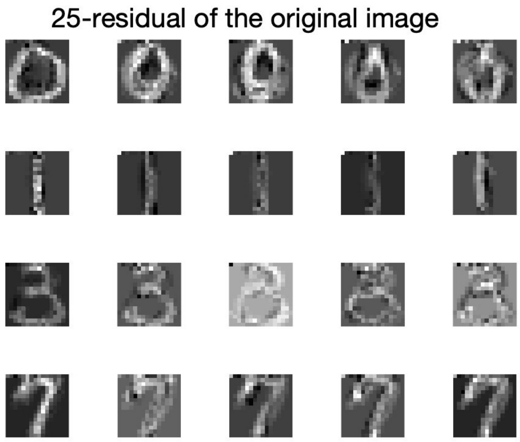
Figure 9: Original data, approximation images and residual between the former and the approximated images by $k = 20$ feature vectors.



(a) Original.



(b) Approximation.



(c) Residual.

Figure 10: Original data, approximation images and residual between the former and the approximated images by $k = 25$ feature vectors.

As we can see from *figure 6* to *figure 10*, as the number of feature vectors increases, more characteristics are visible in the approximation of the images and the pics became more clear. As a consequence, the error of the residual, such as the norm of the difference between the original data vectors and their approximation by k feature vectors, decreases as the number of k increases.

All the previous can be clearly proven by *figure 11*: the plot the norms of the errors as a function of k feature vectors.

Exercise 3

The data file **IrisData.mat** consists of a data matrices X and I. The matrix X consists of 150 vectors, each one having four components. The data correspond to measurements of certain dimensions in three species of flowers, *Iris setosa, Iris versicolor, and Iris* *virginica*, and the components of the data vectors have the following attributes:

$$x = \begin{bmatrix} \text{sepal length in cm} \\ \text{sepal width in cm} \\ \text{petal length in cm} \\ \text{petal width in cm} \end{bmatrix}$$

The data matrix I is the label matrix in which to each column is assigned a species between the three of the flowers.
After I centered the data, I compute the SVD (Singular Value Decomposition) of the centered data matrix and, then, after visualizing the singular values of the diagonal matrix D to see how many feature vectors are needed to approximate the original column vectors in the best way possible, I plot the first 3 components one against the other.
As using the same reasoning applied to the first exercise, there are $\binom{3}{2}$ two-dimensional projections of one component of the principal component against the other.
The use of PCA (Principal Component Analysis) to visualize the matrix's column vectors is very interesting and the reason is that PCA allow the visualisation of centered data in two- or three-dimensional scatter-plot with a the best possible spread. That's because it can be proven that the projection direction that maximizes the spread of the data is given by the first left singular vector, $q = u(1)$:

$$max\{spread(y)\} = \sigma_1 = \|\mathbf{X^T u^{(1)}}\|$$

The other projection direction, orthogonal to u_1, that gives the second largest spread is $u^{(2)}$:

$$max\{spread(\tilde{y})\} = \sigma_2 = \|\tilde{\mathbf{X}}^{\mathbf{T}} \mathbf{u^{(2)}}\|$$

So, let's visualize the scatter-plot of the first 3 principal components, one against the other:

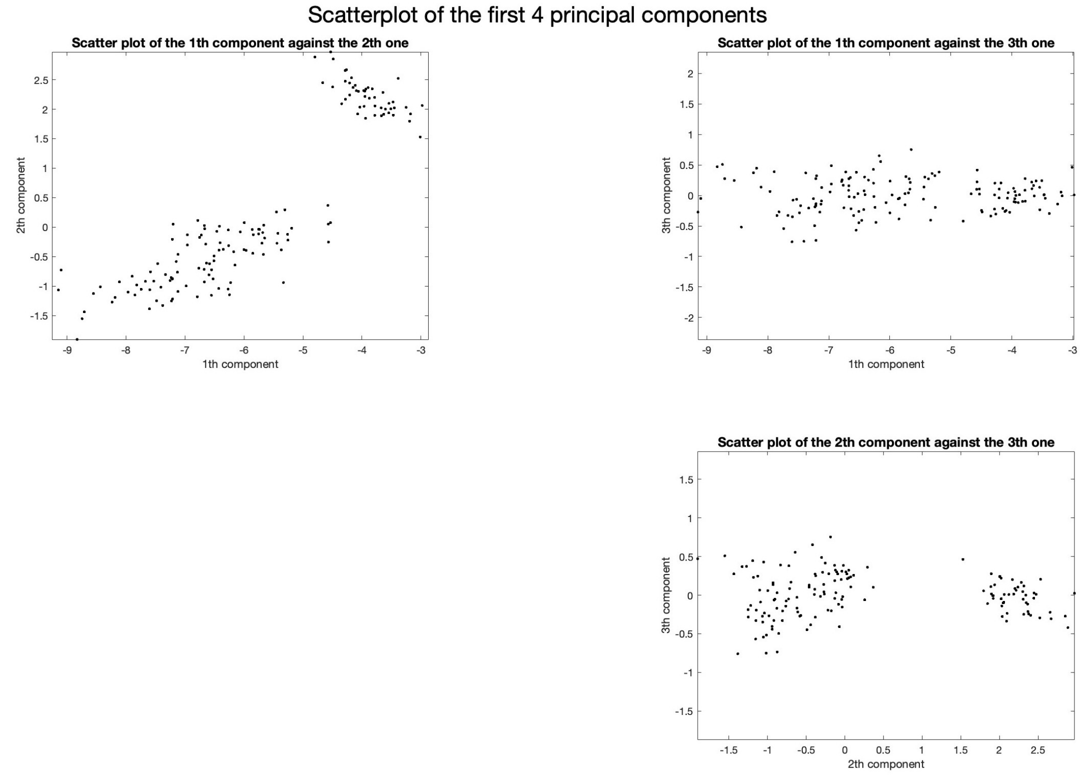


Figure 11: Scatter-plot of the *ith* (with $i = \{1,2,3\}$) against the *jth* one (with $j = \{1,2,3\}$), where i and j represent the columns of the principal component matrix Z_k derived from applying SVD (Singular Value Decomposition) to the centered data matrix X_c of the *IrisDataAnnotated.mat* dataset.

It's interesting to notice the presence of cluster in, approximately, all the scatter-plots and, with a lot of imprecision, we could distinguish the three species of the iris flower in charts of *figure 11*.

Appendices

A. Exercise 1

```
#Part a

clear all
load 'ModelReductionData␣(3).mat'

counter = 0;
figure()
for i = 1:6
    for j = i+1:6
        counter = counter + 1;
        subplot(5, 5, 5*(i-1)+(j-1))
        plot(X(i,:), X(j,:), 'k.', 'MarkerSize', 3)
        axis('equal')
        title("Scatter plot of the " + i + "th component
           against the " + j + "th one", FontSize=8)
        xlabel(i + "th component", FontSize=8)
        ylabel(j + "th component", FontSize=8)
    end
    sgtitle("Scatterplot of one component against the other", '
       FontSize', 25)
end


# Part b

% Center the data and compute the SVD. Plot the singular values
   .
tot = 4000*6;
x_bar = (1/tot)*sum(X,2);
x_c = X  - x_bar;
[U,D,V] = svd(x_c);
figure()
semilogy(diag(D), 'k-*', 'MarkerSize', 10)
set(gca, 'FontSize', 15)
ylabel("Singular value")
xlabel("X's column")
title("Plot of the singular values", FontSize=20)
subtitle("Singular values plotted in a logarithmic scale",
   FontSize=12)


%Show the scatter plots of the first few (I show the first 4)
   principal components.
```

```
Z = U(:,1:4)'*x_c;

figure()
for i = 1:4
    for j = i+1:4
        subplot(3, 3, 3*(i-1)+(j-1))
        plot(Z(i,:), Z(j,:), 'k.', 'MarkerSize', 7)
        axis('equal')
        title("Scatter plot of the " + i + "th component
           against the " + j + "th one", FontSize=13)
        xlabel(i + "th component", FontSize=11)
        ylabel(j + "th component", FontSize=11)
        sgt = sgtitle("Scatterplot of the first 4 principal
           components");
        sgt.FontSize = 22;
    end
end
```

B. Exercise 2

```
clear all
load HandwrittenDigits.mat

% Extract from X the images that correspond to numbers 0, 1, 3
   and 7.
for j = 1:10
    if (j == 1 || j == 2 || j == 4 || j == 8)
        II = find(I == j-1);
        XX = X(:,II(1));
        h = j-1;
        figure()
        imagesc(reshape(XX, 16, 16)')
        colormap(1-gray);
        axis('square')
        axis('off')
        set(gca, 'FontSize', 20)
        title("Images " + h, FontSize=22)
        subtitle("Extracting from the X matrix of the
           HandwrittenDigts dataset an image corresponding to
           the number " + h, FontSize=15)
    end
end

 % I first take 5 samples for each subgroup and plot it in a
    matrix
counter = 0; % I add a counter because I need that the last for
```

```
    loop
            % takes the size of j (so, from 1 to 4) and not
                their actual
            % value (1, 2, 4, 8)
for j = 1:10
    if (j == 1 || j == 2 || j == 4 || j == 8)
        counter = counter + 1;
        count = 0;
        % I first take 5 samples for each subgroup and plot it
            in a matrix
        Ij = find(I == j-1, 5);
        Xj = X(:,Ij(1:5));
        [U,D,V] = svd(Xj);
        for k = [5, 10, 15, 20, 25]
            count = count + 1;
            % Then, I approximate the samples by linear
                combination of the first
            % k feature vectors
            Zj = U(:,1:k)'*Xj(:,1:5);
            Xj1 = U(:,1:k)*Zj;
            %the residual, x(j)      Pkx(j)
            X_res = Xj - Xj1;
            %error norm
            err = norm(X_res);
            for i = 1:5
                %original images
                figure(5)
                subplot(4, 5, 5*(counter-1)+i)
                imagesc(reshape(Xj(:,i), [], 16)')
                colormap(1-gray);
                axis('square')
                axis('off')
                set(gca, 'FontSize', 20)
                sgtitle("5 samples of the images")


                %k-approximation image
                figure(count + 5)
                subplot(4, 5, 5*(counter-1)+i)
                imagesc(reshape(Xj1(:,i), [], 16)')
                colormap(1-gray);
                axis('square')
                axis('off')
                set(gca, 'FontSize', 20)
                sgtitle(k + "-approximation of the original
                    image", fontsize=22)
```

```
                %k-approximation image
                figure(count + 10)
                subplot(4, 5, 5*(counter-1)+i)
                imagesc(reshape(X_res(:,i), [], 16)')
                colormap(1-gray);
                axis('square')
                axis('off')
                set(gca, 'FontSize', 20)
                sgtitle(k + "-residual of the original image",
                    fontsize=22)



            end

        end
    end
end


%norms of the residual in function of k
counter = 0;
k_val = [5, 10, 15, 20, 25];
err = zeros(1,5);
%norm of the residual in function of k
for j = 1:10
    if (j == 1 || j == 2 || j == 4 || j == 8)
        counter = counter + 1;
        err = zeros(5,1);
        count = 0;
        % I first take 5 samples for each subgroup and plot it
            in a matrix
        Ij = find(I == j-1, 5);
        Xj = X(:,Ij(1:5));
        [U,D,V] = svd(Xj);
        for k = [5, 10, 15, 20, 25]
            count = count + 1;
            err(count,:) = zeros(1,5);
            % Then, I approximate the samples by linear
                combination of the first
            % k feature vectors
            Zj = U(:,1:k)'*Xj(:,1:5);
            Xj1 = U(:,1:k)*Zj;
            %the residual, x(j)     Pkx(j)
            X_res = Xj - Xj1;
            err(count) = norm(X_res);
            %error norm
        end
```

```
        figure(count + 15)
        semilogy(k_val, err, '*-')
        grid on
        title("norm of the residual in function of " + k)
    end
end
```

C. Exercise 3

```
clear all
load IrisDataAnnotated.mat
% By using the PCA, investigate if the data set suggests the
    presence of
% clusters that would make it possible to separate the three
    species from
% each other.
tot = 150*4;
x_bar = (1/tot)*sum(X,2);
x_c = X  - x_bar;
[U,D,V] = svd(x_c);
figure()
semilogy(diag(D), 'k-*', 'MarkerSize', 10)
set(gca, 'FontSize', 15)

Z = U(:,1:4)'*x_c;
figure()
for i = 1:3
    for j = i+1:3
        subplot(2, 2, 2*(i-1)+(j-1))
        plot(Z(i,:), Z(j,:), 'k.', 'MarkerSize', 7)
        axis('equal')
        title("Scatter plot of the " + i + "th component
            against the " + j + "th one", FontSize=13)
        xlabel(i + "th component", FontSize=11)
        ylabel(j + "th component", FontSize=11)
        sgt = sgtitle("Scatterplot of the first 4 principal
            components");
        sgt.FontSize = 22;
    end
end
```